Assessment Task 3: Visual Analytics -

Australian International Trade dataset

ZOOM in for better view

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Introduction

This report utilizes the provided Australian International Trade Dataset which can be found in the Australian Bureau of Statistics (Australian Bureau of Statistics, 2024). The dataset contains 36 years' worth of data shown in dollar values (million) for 10 main categories, and a total of 67-sub categories, including each year's trade total worth for all the categories combined. The purpose of this report is to convey the findings of the dataset to stakeholders – individuals within public policy circles or government, who are interested in understanding the Australia's international trade trends from 1988 to 2024. The analysis applies statistical, analytical and raw-dollar patterns on the import and export data. These allow the possibility of uncovering long-term shifts across categories. Through visualisations developed in Tableau, the report compares all main product categories, identifies significant trends and change points, and explores in detail the performance of a selected subcategory. The tools used for this report include Excel for raw data pattern conversions, category calculations, and value formatting, then the use of tableau, which moulds the datasheets into meaning visualisations that allow the creation of dashboards and storyboards.

Dataset Description

The export and import data provided on separate sheets consisted of trade values in Australian dollars (AUD) in millions, along with technical metadata such as Series ID, Units, and Export/Import labels. It also displayed the main category and subcategories in the format in which the first column for that main category and year had the sum of its total values for the subsequent subcategories. An example is shown below:

M	N	O
1 Beverages and to	11 Beverages	12 Tobacco and tob
$ Millions	$ Millions	$ Millions
A1827815T	A1827816V	A1827817W
273	250	22

Data Transformations

In order to supply tableau with formatted data, a series of transformations have been completed on the excel workbook- specifically on the export and import sheets. The first step was to create analytical and statistical data patterns for data visualisation. Establishing these data relationships and data patterns using data queries can bring out interactive visualisations that explain insight stories about the data.

Sheets created: each box is a sheet

Export-analytical	Import-analytical

Analytical Combined

Used to calculate the year-on-year percentage change in trade value for each category or subcategory.

General formula applied across rows:

Example for Import Analytical:

=IFERROR(INDIRECT(ADDRESS(ROW()+1, COLUMN(), 4, TRUE, "Export-dataset")) / INDIRECT(ADDRESS(ROW(), COLUMN(), 4, TRUE, "Export-dataset")), "")

This calculates the percentage change from the previous year to the current year for each category.

Export-statistical	Import-statistical

Statistical Combined

Used to calculate the percentage contribution of each category or subcategory to its total.

Formulas used:

=B4 / $CA4

calculates the percentage of a main category (e.g., 0.Food) relative to the total export or import value for that year (in column CA).

=C4 / $B4

=D4 / $B4 .... So on

calculate the percentage contribution of each subcategory under Category 0 to its own main category total.

Dollar combined

No formulas used here, just using original export and import data raw dollar values for export and import into one sheet.

Now since all the export and import data is combined for these 3 finalised sheets, there will be duplicate dates, so in order to differentiate between which dates are export and which are import, a new column was inserted after the date column called trade. This had the values ‘export’ for the matching export cells and ‘import’ for its following import cell. So now the data for each of these sheets is stacked (i.e 1988-2024 export ontop, then continuing with 1988-2024 import data). The flexibility with tableau is that once these sheets are uploaded, each sheet can individually be filtered based on the trade category.

Also, these new sheets had the meta data such as series id, and $ Millions removed for a simpler datasheet, as they provide no valuable use for the visualisations. Another change is that in the analytical combined sheet the year 1988 was removed as you cannot calculate a change for 1988, because there's no 1987 data.

Main Category Visualisations

Raw-dollar value trends

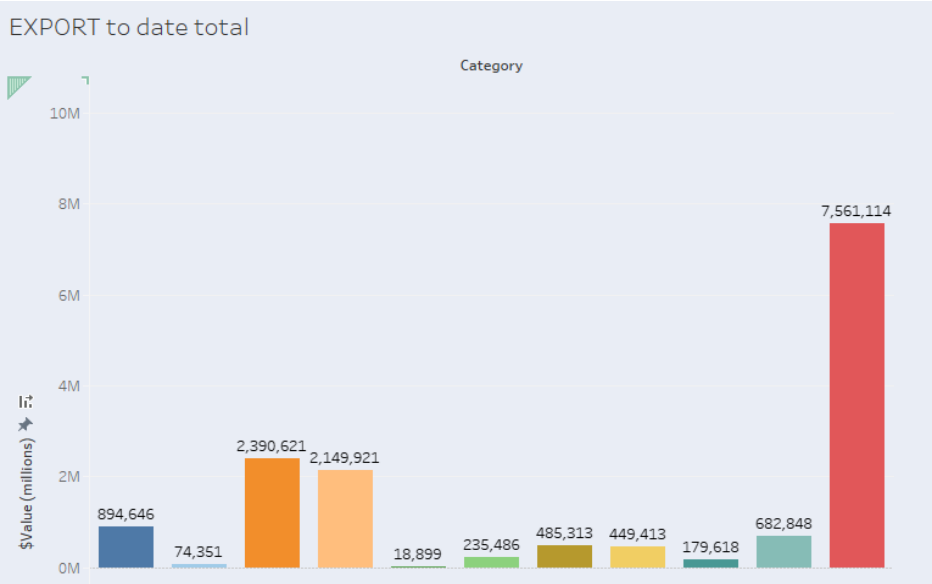


Figure 1: Bar chart displaying total export (1988-2024) of different categories

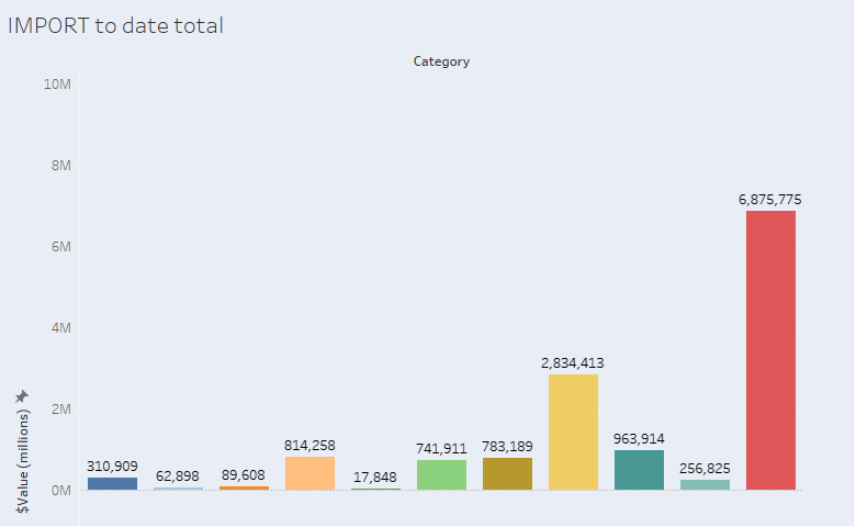


Figure 2: Bar chart displaying total import (1988-2024) of different categories

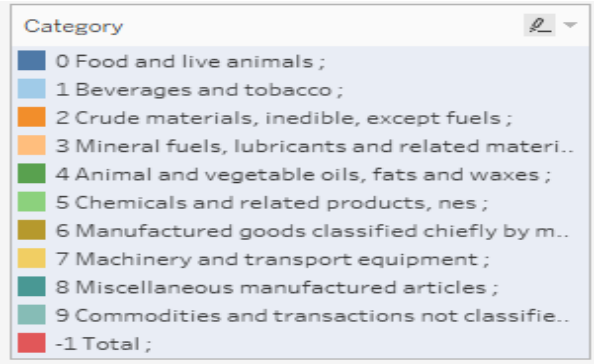


Figure 3: Key for the categories presented in figures 1 and 2

The above graphs show the cumulative import and export dollar values in millions from 1988 to 2024. According to figure 1, Australia earns most from category 2 -Crude materials and category 3 -mineral fuels with each above 2 trillion dollars. According to figure 2, Australia imports much less of its 2 highest exported categories, meaning that there may be an abundance of those categories present already. The lowest of both exports an imports is category 4- Animal and vegetable oil, gaining around 18.9 billion in exports, and spending 17.8 billion in imports. It is also seen that Australia has highest imports in category 7- machinery and transport where “Primary exporters to Australia include China, the United States, and Japan” (Connecta Network, 2024). The colours used for each graph on the categories were consistant in both figures 1 and 2, to show a clear comparison between import values and export values, and the use of labels fro each category helps the reader to make quantitative comparisons.

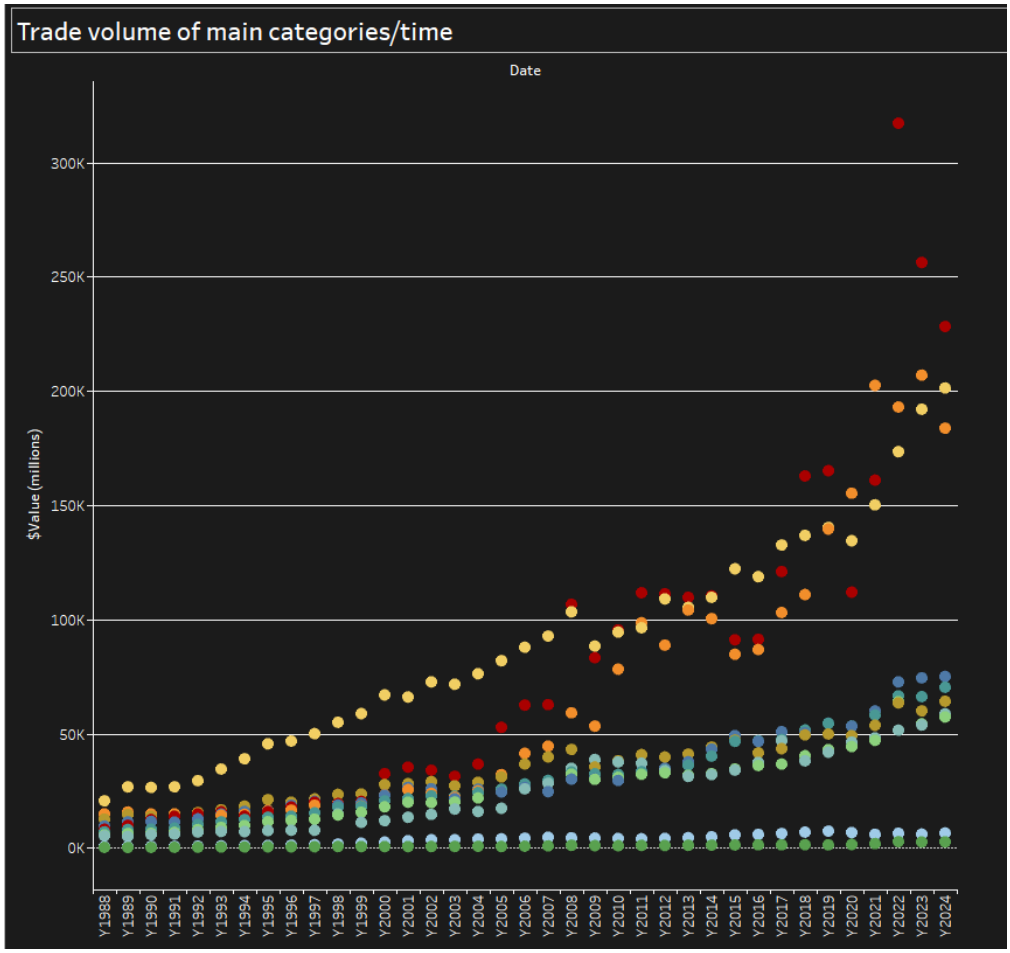


Figure 4: Dot plot of total import and export (volume) of main categories over time.

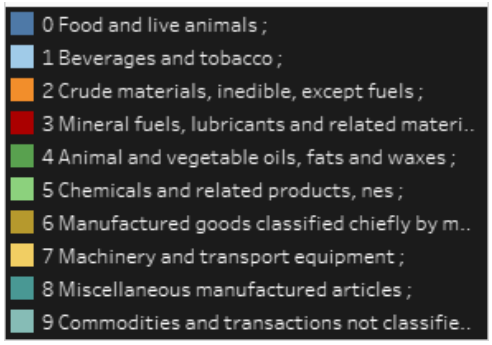


Figure 5: Key for figure 4

Figure 4 presents the total trade value for a specific category in a specific year. This value combines both the export and import per category. As seen in figure 4, there is a steep upward trends: Categories 2 (Crude materials) 3, (Mineral fuels) and 7 (Mineral fuels) show strong, consistent growth, with Mineral Fuels having an excessive outlier of over 300 000 million (AUD). This aligns with Australia's global role in exporting raw materials and fuels. Categories 0, 5, 6, 8, 9 show s show a steady increase which shows that these sectors support a balanced trade mix, which is influenced by consumer demand and global supply chains. Categories like 1 (Beverages), 4 (Animal/vegetable oils), and 5 (Chemicals) remain relatively small contributors. The colours used for the plots along with the type of chart display a clear story of how certain categories excel, while others stay steady, and others are neutral, with the help of the legend provided in figure 5. The contrasting black background creates a clear distinction between these categories.

Statistical Pattern

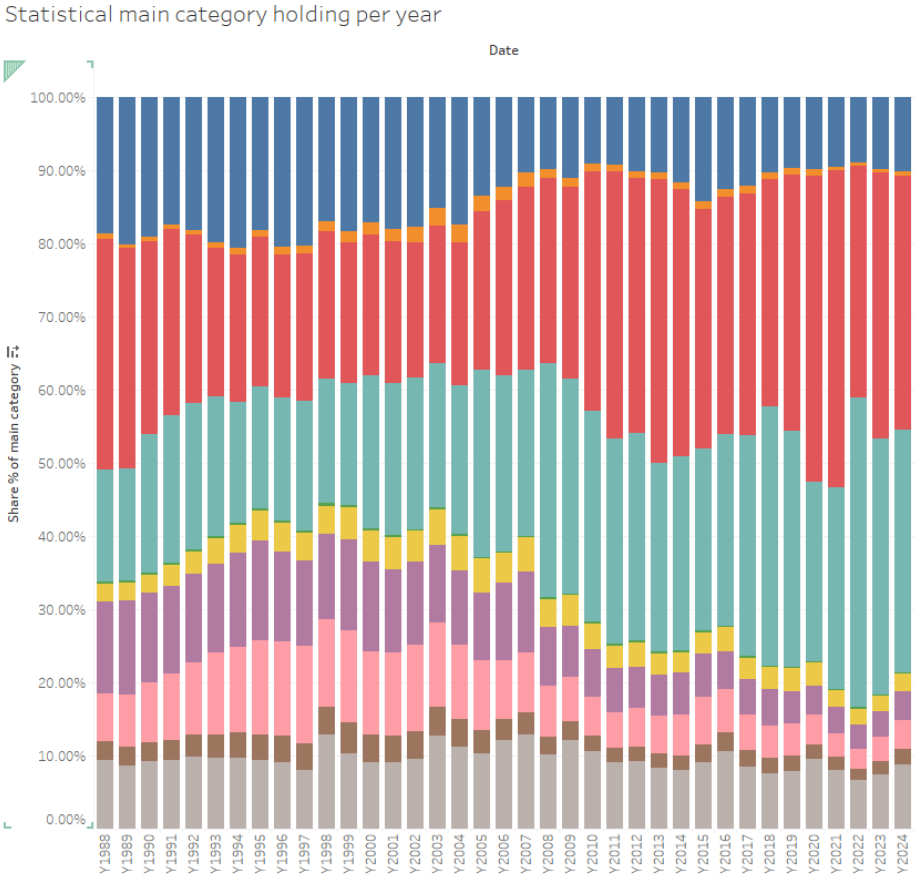


Figure 6: Stacked bar chart showing how much % of total exports each category holds per year

Figure 7: legend for figure 6

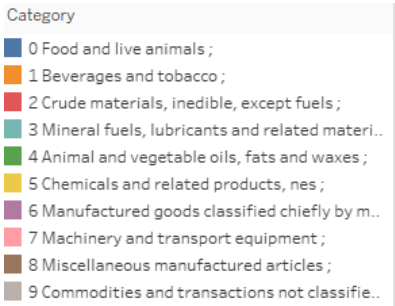
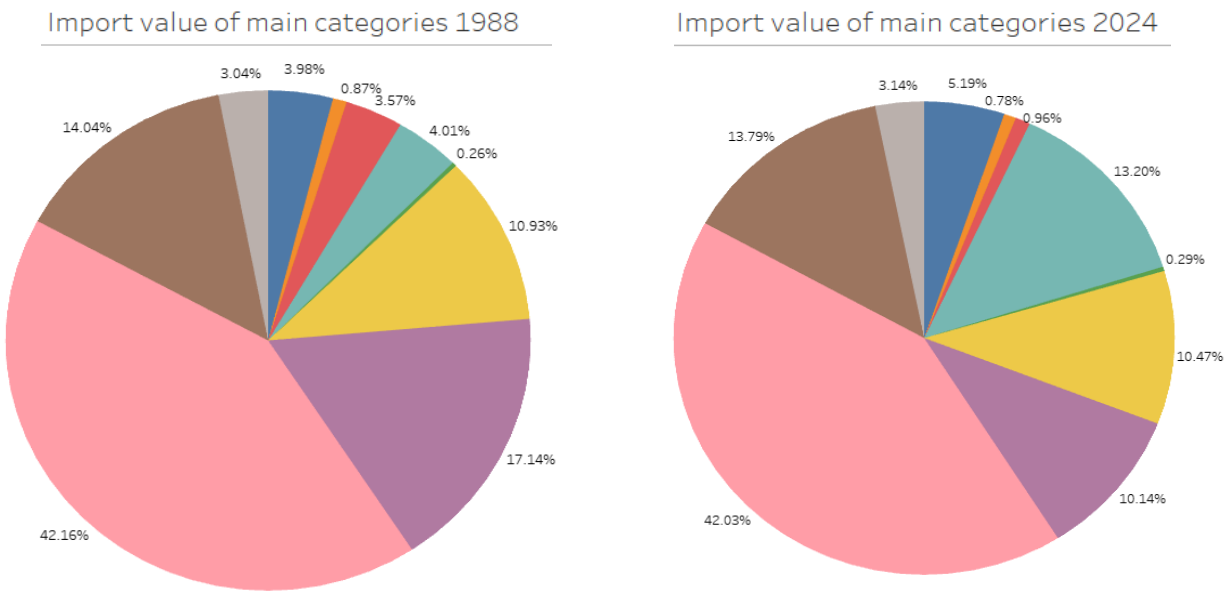


Figure 6 presents a stacked bar chart that visualises the $ share of each main export category. Each bar consists of one years entire trade composition. The colours used were to differentiate between the category, as supported in figure 7. Key observations include a dominance in category 2 crude materials, where the red is starting to get bigger overtime, and category 3 mineral fuels, also gaining value in the recent years. Although this doesn't show any values, it is a perfect visual fact provider to understand the time series of export of the main categories.

Figure 8 Pie graphs of total import composition 1988 (first pie) and 2024 (second pie):



These pie graphs are essential in comparing and drawing connections to Australian imports in 1988, and in 2024. As can be seen above in figure 8, the import rates of categories in much different time periods are seen to be almost exactly the same, showing that Australia invests uniformly the same amount of money to bring products from these main categories to Australia. Some major differences include the increase of Mineral Fuel imports as Exports of liquefied natural gas have increased an average of 13 per cent a year over the past decade (Australian Government, 2023). Again, the colours used in both pie charts was enough to contrast between categories, and the side by side comparison alongside the labels is enough for the readers to understand the story of how these categories have changed/stayed consistent over long periods of time.

Analytical Pattern

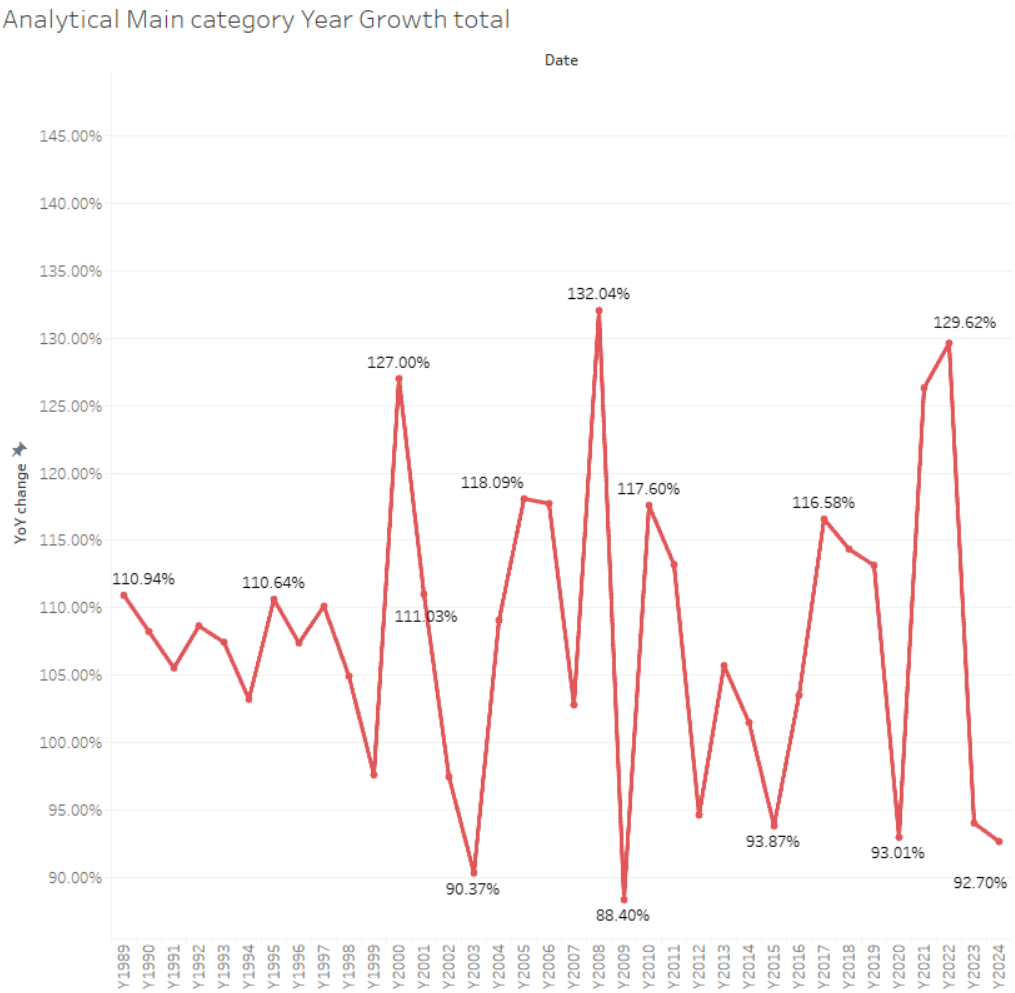


Figure 9: Line chart visualising the year-over-year (YoY) percentage change in total export value across all main categories (i.e. -1 TOTAL ;) from 1989 to 2024.

This chart summarises the volatility and resilience of Australia's total exports over 35 years using the -1 TOTAL field from the Export Analytical dataset. The data reveals substantial growth most notably in 2021, which saw a 35.8% increase, likely reflecting strong post-COVID recovery across export sectors (*Australia—Services Exports Rising Gradually Following Border Reopening*, n.d.). Other high-growth years include 2010 (+33.0%), and 2000 (+30.1%). In contrast, a significant decline occurred in 2009 (−33.1%) during the Global Financial Crisis, however quickly rebounded as "Australia's economy was buoyed by large resource exports to China" (Reserve Bank of Australia, 2025). Smaller but still notable drops were recorded in 2001, 2002, and 2012, each exceeding a 12% decrease. These fluctuations depict the volatility of Australia's export economy in response to global events.

Relationships between Main categories and Subcategories - FOCUS ON CATEGORY 0

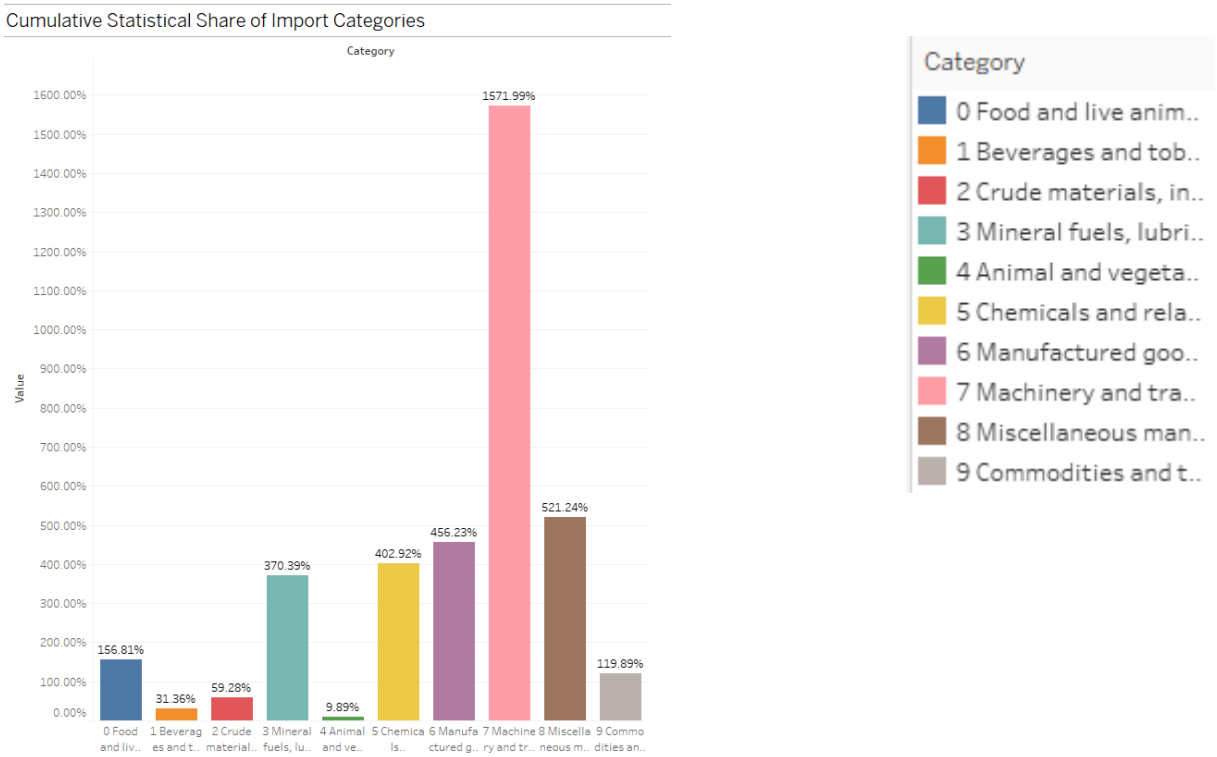


Figure 10: showing cumulation of total import percentage share of a main category across all 36 years

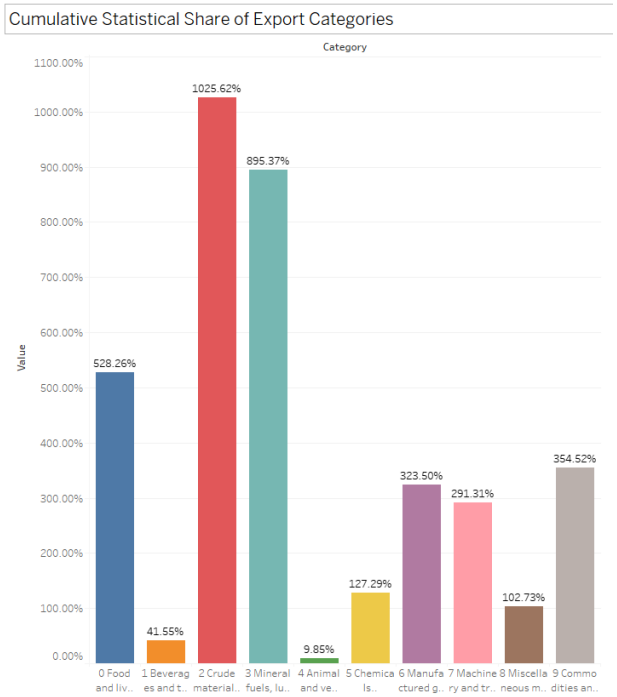
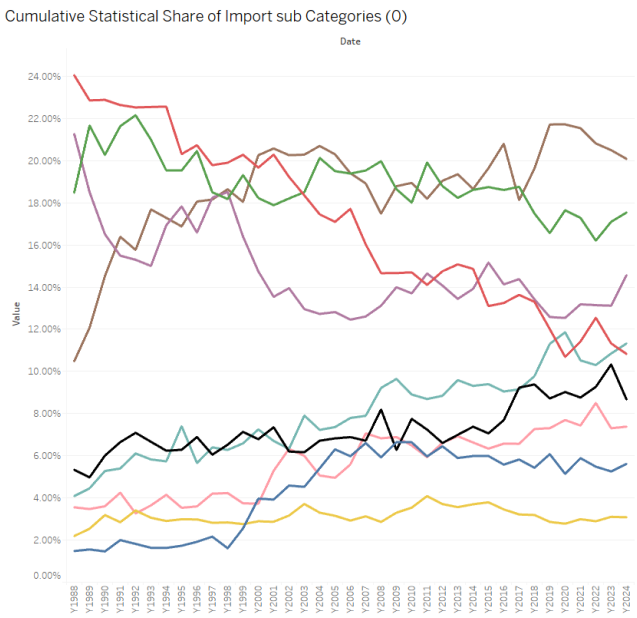
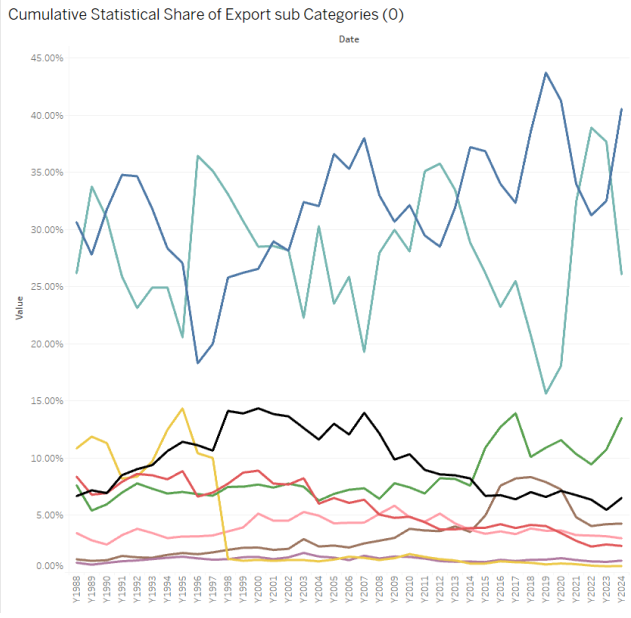
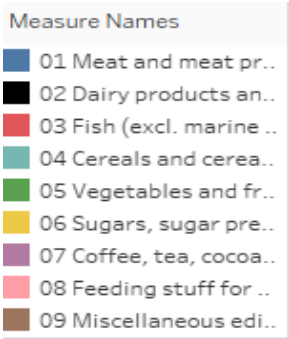


Figure 11: showing cumulation of total export percentage share of a main category across all 36 years







Figures 12 and 13 (left to right): Import and Export subcategories (respectively). Line graphs that track the percentage share of each subcategory under Category 0. Figure 12 links to figure 10 (import) and figure 13 links to figure 11 (export)

Figures 10 and 11 provide an overview of the cumulative statistical share of all main import and export categories between 1988 and 2024. While Category 0 (Food and live animals) shows moderate cumulative growth in imports (156.81%) and strong performance in exports (528.26%), it is not the leading category in either. However, the subcategory analysis in Figures 12 and 13 reveals the underlying drivers of these trends.

In Figure 12, which illustrates the import shares of Category 0 subcategories, the share of imports has become more evenly distributed over time. Notably, the contribution of 01 Meat declined significantly from over 24% to under 12% by 2024, while subcategories like 05 Vegetables and 06 Sugars increased steadily, suggesting changes in domestic consumption patterns or local production capabilities. This may reflect improvements in Australia's agricultural self-sufficiency or trade diversification.

Figure 13, shows that Australia's food export strength is concentrated in a few key subcategories. 01 Meat and 02 Dairy products together often make up over 60–70% of Category 0 exports, though their values fluctuate year to year. The sharp volatility in these subcategories may be due to external shocks such as droughts, livestock disease outbreaks, or global trade demand

Together, the figures show that although Category 0 is not Australia’s largest trade category overall, it plays a strategically important role, especially in exports with particular subcategories such as meat and dairy serving as core contributors. This analysis highlights the importance of tracking both category-level growth and subcategory trends to understand structural changes in Australia's trade economy.

Sub-Category focus (01 Meat and meat preparations)

Blue- export, Orange- import

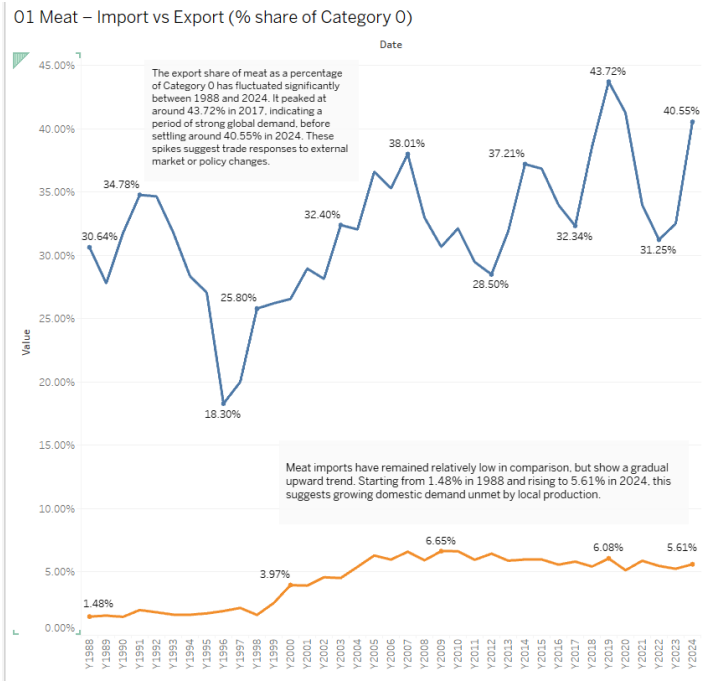


Figure 14-line chart comparing the percentage share of meat products (01) in relation to the total value of Category 0 (Food and Live Animals)

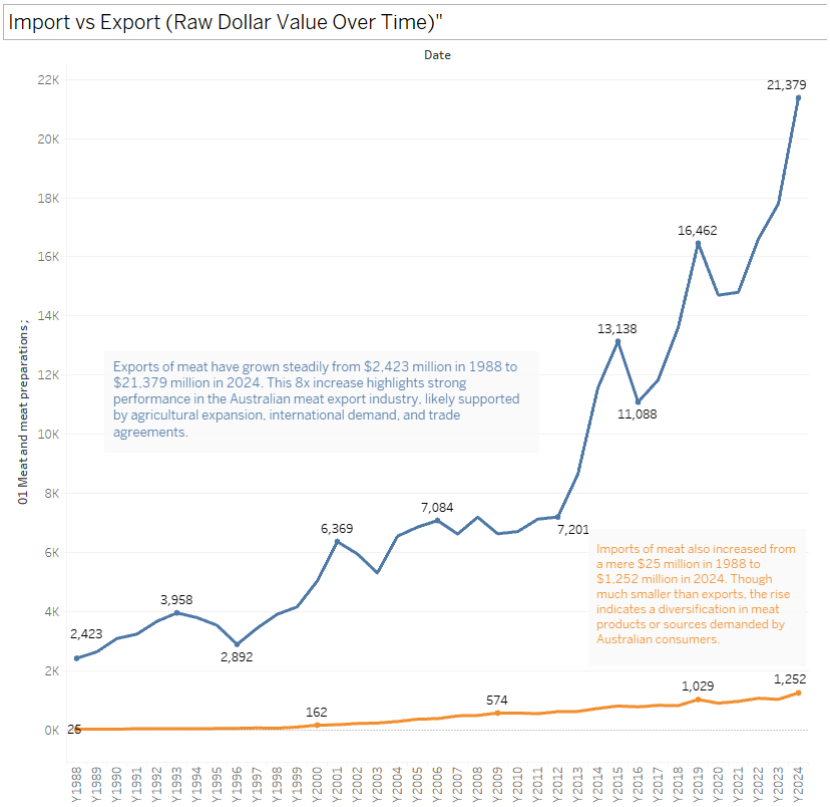


Figure 15-line chart displaying the actual trade values (in millions) of meat imports and exports from 1988 to 2024.

The visualisations in figures 14 and 15 compare the export and import trends of meat products (01), sub-category of 0 -Food and live animals. The first visualisation shows the percentage share of meat relative to all other subcategories within Category 0 (statistical trend), while the second illustrates the raw dollar value over time (actual trade volume). In both cases, exports vastly outweigh imports, reflecting Australia's strong position as a major meat exporter. The export share has shown fluctuations, potentially tied to global demand shocks, policy changes, or supply chain disruptions. Meanwhile, imports have risen steadily, possibly driven by consumer preference for variety or foreign meat products. Together, these visuals highlight how Australia's meat industry is export-dominant, but also increasingly reliant on certain imports, which could be critical for trade planning and food security strategies. The visualisations utilize area annotations to point out key labels, which provide reasoning and explaining trends for successful tory telling. The colour scheme is also consistent with blue being export and orange being import.

2015 – 2024 Trend

Import of (01 Meat and meat preparations)

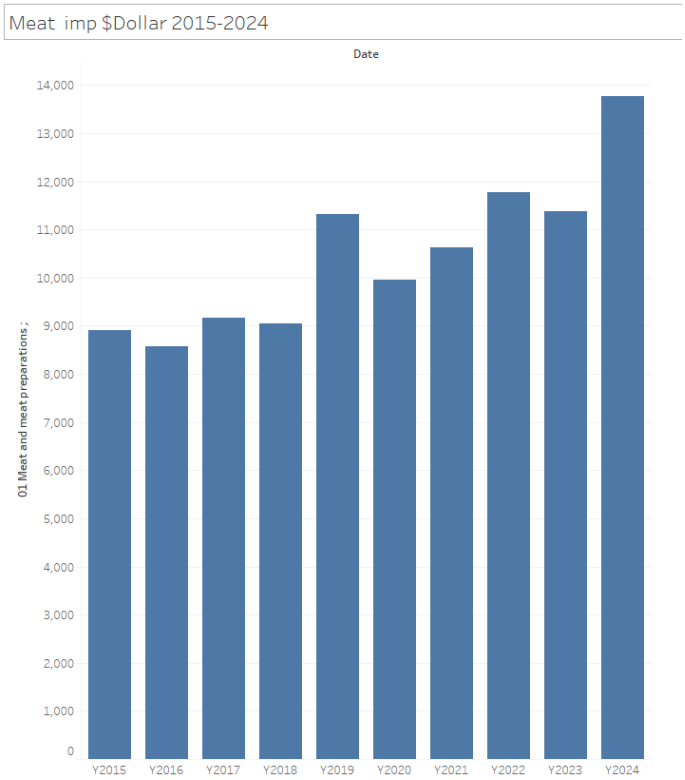
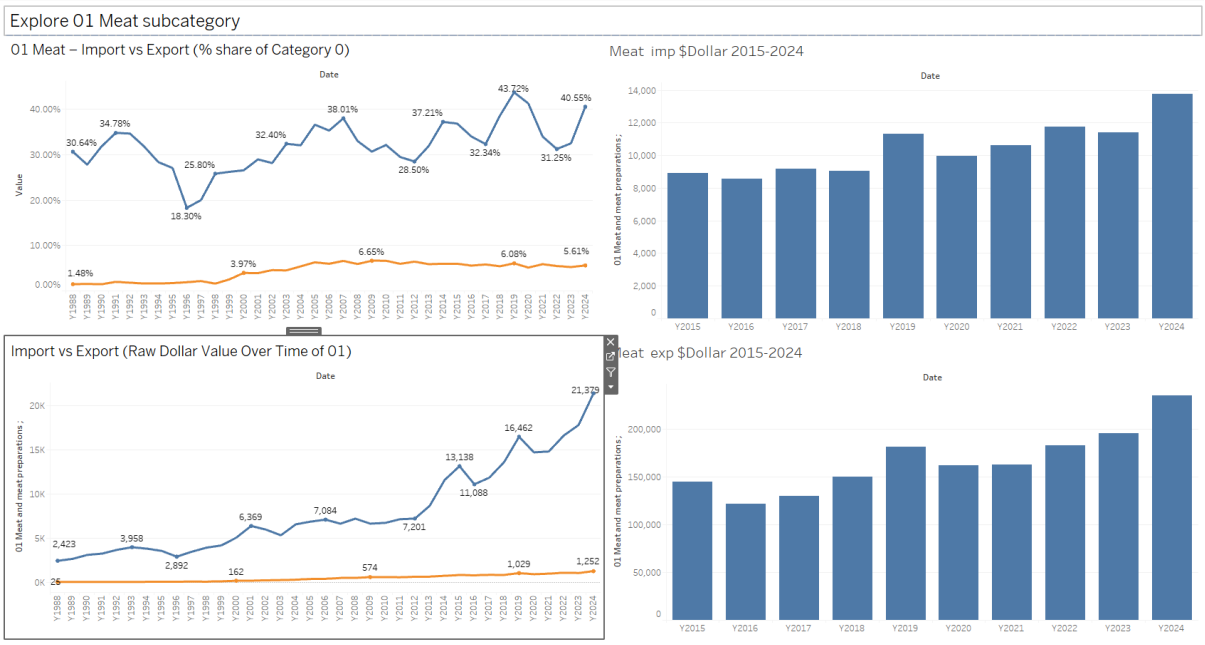


Figure 16 Import values from 2015 – 2024

Between 2015 and 2024, the import value of Meat and meat preparations (01 category) shows a steady upward trend. The value starts at approximately $8,900 million AUD in 2015 and rises to over $13,800 million AUD by 2024, representing a gain of around 55%

over the decade. This growth suggests increasing domestic demand for meat products that cannot be met by local supply, possibly driven by: Population growth, Changes in dietary habits, And seasonal or economic variations in local production. There's a minor plateau around 2016–2018, but a strong acceleration after 2019. Based on the upward trend in meat imports from 2015 to 2024, continued growth is expected over the next five years unless disrupted by factors such as trade agreement changes, improved local production, or increased import tariffs. To manage this, Australia should diversify its sourcing countries to reduce risk, invest in local meat processing as a form of import substitution, and monitor global meat prices to adjust targets accordingly. If domestic supply remains limited, a 5–7% annual import growth target is reasonable; however, if substitution efforts succeed, import growth should be capped at 2–3% to reduce dependency.

Story Board



Summary and Conclusion

This report has successfully explored Australia's international trade dynamics across 36 years, using a multi-layered analytical approach applied to the Australian International Trade Dataset. Through a combination of raw-dollar, statistical, and analytical visualisations, it uncovered key insights across all main product categories, while delivering a focused case study on the subcategory 01 Meat and meat preparations under Category 0 – Food and Live Animals.

The designed graphs clearly illustrated Australia’s dominance in the export of categories like Mineral Fuels and Crude Materials, and its heavy reliance on imported machinery. In the subcategory analysis, the percentage share and trade volumes of 01 Meat were examined over time, revealing a long-term export surplus and steady import growth, suggesting Australia’s dual role as a meat supplier and consumer of imported varieties. The comparison between subcategory and main category trends highlighted strategic trade shifts and helped identify change points driven by both global demand and domestic capacity.

A ten-year trend analysis (2015–2024) for meat imports showed a 55% rise in trade value, pointing to increased consumption or supply shortfalls. Based on this, a forward looking strategy was proposed for 2025–2025, aiming for a 5–7% growth if imports are necessary, or capping at 2–3% if domestic production can substitute supply.

The dashboards created in Tableau were key in showing 01 Meat export/import trades. They helped uncover trends and enabled clear side-by-side comparison of statistics for that sub category. Overall, the report demonstrates how visual analytics can be used to support informed decision-making for public policy, international trade strategies, and long-term economic planning.

References

Australia—Services exports rising gradually following border reopening. (n.d.). Www.exportfinance.gov.au. https://www.exportfinance.gov.au/resources/world-risk-developments/2022/august/australia-services-exports-rising-gradually-following-border-reopening/Australian Bureau of Statistics. (2024). International trade | Australian Bureau of Statistics. Www.abs.gov.au. https://www.abs.gov.au/statistics/economy/international-tradeAustralian Government. (2023, November 30). Energy trade | energy.gov.au. Energy.gov.au. https://www.energy.gov.au/energy-data/australian-energy-statistics/energy-tradeConnecta Network. (2024, August 22). What are Australia's Main Exports and Imports? Connecta Network | Freight Forwarders Network; Connecta. https://www.connecta-network.com/what-are-australias-main-exports-and-imports/Reserve Bank of Australia. (2025). The Global Financial Crisis. Reserve Bank of Australia; Reserve Bank of Australia. https://www.rba.gov.au/education/resources/explainers/the-global-financial-crisis.html